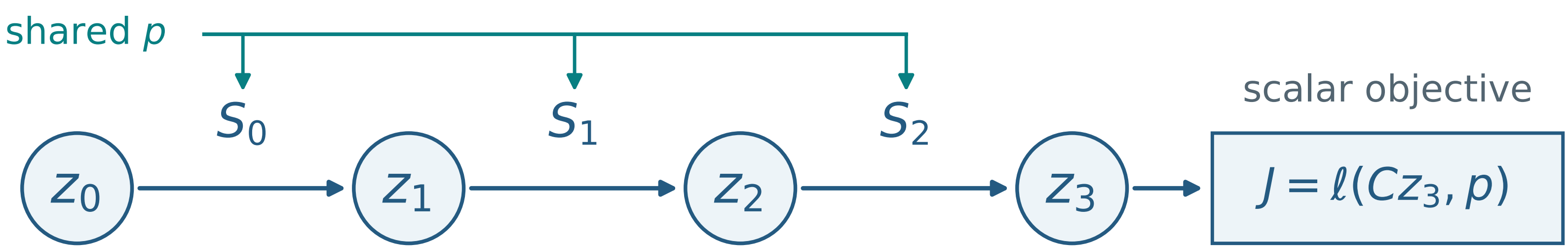
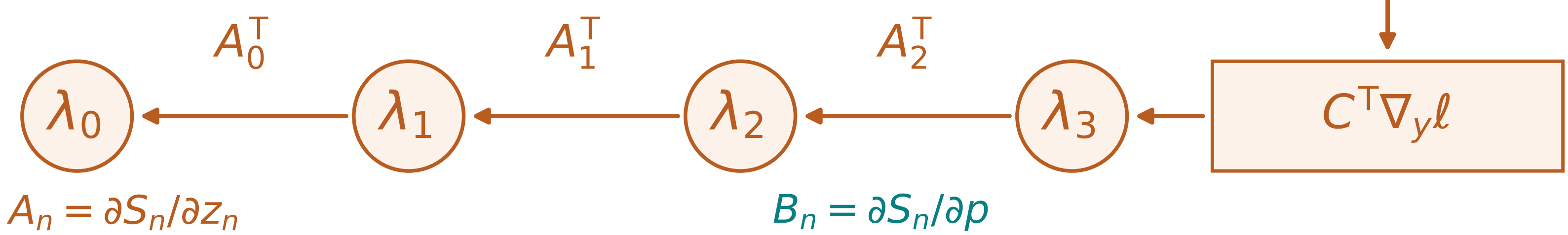
Backpropagation through three updates

1 Evolve the state

$$z_{n+1} = S_n(z_n, p)$$



2 Pass adjoints backwards



3 Add every parameter contribution

from updates $B_0^\top \lambda_1 + B_1^\top \lambda_2 + B_2^\top \lambda_3$

$$\frac{dJ}{dp} = \underbrace{\frac{\partial \ell}{\partial p}}_{\text{direct loss}} + \underbrace{\sum_{n=0}^{2} B_n^\top \lambda_{n+1}}_{\text{all three updates}} + \underbrace{\left(\frac{\partial z_0}{\partial p}\right)^\top \lambda_0}_{\text{initial state}}$$

The gradient is an input to a separate optimisation step.